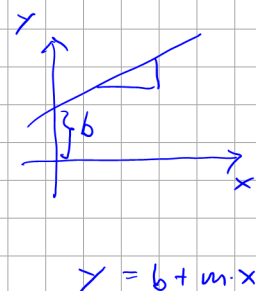
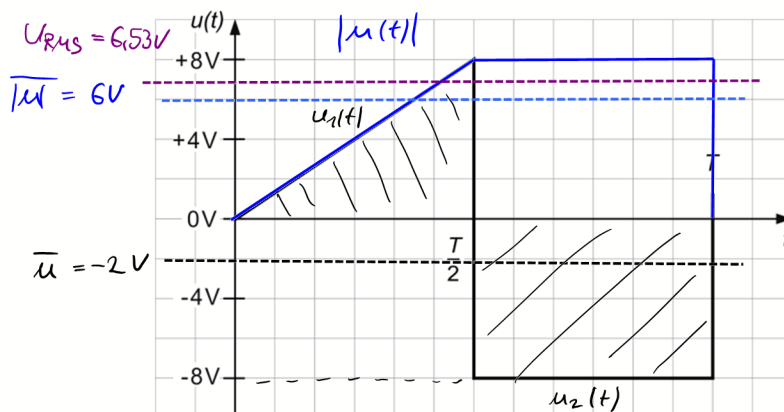


### 1.3 Übungsaufgaben zu Kapitel 1

#### PER 1

Gegeben ist eine Wechselspannung:



Berechnen Sie den

- Gleichwert (Mittelwert)
- Gleichrichtwert
- Effektivwert.

$$\bar{u} \leq \overline{|u|} \leq U_{RMS}$$

a)

$$\bar{u} = \frac{1}{T} \int_0^T u(t) dt = \frac{1}{T} \left[ \int_0^{\frac{T}{2}} u_1(t) dt + \int_{\frac{T}{2}}^T u_2(t) dt \right]$$

Fläche

$$= \frac{1}{T} \left[ \frac{T}{2} \cdot 8V \cdot \frac{1}{2} + \frac{T}{2} (-8V) \right]$$

$$= \frac{1}{T} [2VT - 4VT]$$

$$= 2V - 4V = \underline{\underline{-2V}}$$

b)

$$\overline{|u|} = \frac{1}{T} \int_0^T |u(t)| dt = \frac{1}{T} \left[ \int_0^{\frac{T}{2}} |u_1(t)| dt + \int_{\frac{T}{2}}^T |u_2(t)| dt \right]$$

$$= \frac{1}{T} \left[ \frac{T}{2} \cdot 8V \cdot \frac{1}{2} + \frac{T}{2} \cdot 8V \right]$$

$$= 2V + 4V = \underline{\underline{+6V}}$$

$$\begin{aligned}
 c) \quad U_{RMS} &= \sqrt{\frac{1}{T} \int_0^T u^2(t) dt} \\
 &= \sqrt{\frac{1}{T} \left[ \int_0^{\frac{T}{2}} u_1^2(t) dt + \int_{\frac{T}{2}}^T u_2^2(t) dt \right]} \\
 &= \sqrt{\frac{1}{T} \left[ \int_0^{\frac{T}{2}} \left( \frac{8V}{\frac{T}{2}} \cdot t \right)^2 dt + \int_{\frac{T}{2}}^T (-8V)^2 dt \right]} \\
 &= \sqrt{\frac{1}{T} \left[ \int_0^{\frac{T}{2}} \left( \frac{16V}{T} \right)^2 t^2 dt + \int_{\frac{T}{2}}^T (8V)^2 dt \right]} \quad 1=t^0 \\
 &= \sqrt{\frac{1}{T} \left[ \left( \frac{16V}{T} \right)^2 \frac{t^3}{3} \Big|_0^{\frac{T}{2}} + (8V)^2 t \Big|_{\frac{T}{2}}^T \right]} \\
 &= \sqrt{\frac{1}{T} \left[ \frac{(16V)^2}{T^2} \frac{T^3}{8 \cdot 3} + (8V)^2 \cdot \left( T - \frac{T}{2} \right) \right]} \\
 &= \sqrt{\frac{16^2 V^2}{24} + 8^2 V^2 \cdot \frac{1}{2}} \\
 &= \underline{\underline{6,53 V}}
 \end{aligned}$$





